

Crossed-Null Chronometry: Priority Audit and Dynamical Viability Test

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Archive reconstruction from the surviving project ledger. Historical sandbox binaries were ephemeral;
see RECOVERY_PROVENANCE.md.

7. Failure of the exact-null action

The first complete constrained action was

$$S_0 = \int d^4x \sqrt{-g} \left[\frac{M_{\text{P}}^2}{2} R + A_+ Y_+ + A_- Y_- - \frac{\kappa}{2} (C - \sigma^2)^2 - \frac{1}{2} (\nabla \sigma)^2 - V(\sigma) \right],$$

where

$$Y_{\pm} = (\nabla \phi_{\pm})^2, \quad C = -\frac{1}{2} \nabla \phi_+ \cdot \nabla \phi_-.$$

Around

$$\bar{\phi}_+ = m(t - x), \quad \bar{\phi}_- = m(t + x),$$

the linearised constraints are

$$(\partial_t + \partial_x) \pi_+ = 0, \quad (\partial_t - \partial_x) \pi_- = 0.$$

Define

$$s = \frac{\pi_+ + \pi_-}{\sqrt{2}}, \quad r = \frac{\pi_+ - \pi_-}{\sqrt{2}}.$$

The lock fluctuation is

$$\delta C = \frac{m}{\sqrt{2}} (\dot{s} - \partial_x r).$$

Neither the linear constraints nor the quadratic lock contains ∂_y or ∂_z . Transverse gradients first enter nonlinearly through terms such as $(\nabla \pi_{\pm})^2$. The quadratic principal symbol is therefore rank deficient for generic transverse perturbations. Nonlinear terms control directions with no quadratic restoring operator.

Verdict: the exact-null constrained action is not a controlled ordinary 3 + 1D EFT.

The positive homogeneous Hessian did not detect this because it tested only algebraic order-parameter fluctuations.

8. Healthy soft-null repair

The minimal repair makes the phases ordinary propagating compact fields and treats crossed nullity as a background state:

$$\mathcal{L}_{\text{EFT}} = -\frac{F}{2}(Y_+ + Y_-) + \frac{\kappa}{2}(C - m^2)^2.$$

Define

$$a = \frac{\kappa m^2}{2}, \quad \epsilon = \frac{a}{F}.$$

The quadratic Lagrangian in s, r is

$$\begin{aligned} \mathcal{L}^{(2)} = & \frac{1}{2}(F + a)\dot{s}^2 + \frac{1}{2}F\dot{r}^2 - a\dot{s}\partial_x r \\ & - \frac{F}{2}[(\nabla s)^2 + (\nabla r)^2] + \frac{a}{2}(\partial_x r)^2. \end{aligned}$$

The Hamiltonian is positive when

$$F > 0, \quad 0 < a < F,$$

or $0 < \epsilon < 1$. The principal polynomial factorises:

$$F(\omega^2 - k^2) [(F + a)\omega^2 - Fk^2 + ak_x^2] = 0.$$

The modes are

$$\omega_1^2 = k^2,$$

$$\omega_2^2 = \frac{Fk^2 - ak_x^2}{F + a}.$$

Writing $k_x = k \cos \theta$,

$$c_2^2(\theta) = \frac{1 - \epsilon \cos^2 \theta}{1 + \epsilon}.$$

For $0 < \epsilon < 1$, the second mode is real, positive, and subluminal for every direction. The model belongs to the established two-superfluid/entrainment class rather than defining a new species of matter [Superfluid2015].

Local mediator completion

Introduce a healthy massive scalar H :

$$\mathcal{L}_{\text{med}} = -\frac{F}{2}(Y_+ + Y_-) - \frac{1}{2}(\nabla H)^2 - \frac{1}{2}M^2 H^2 + g_H H(C - m^2).$$

Integrating it out below M gives

$$\Delta\mathcal{L} = \frac{g_H^2}{2}(C - m^2)\frac{1}{M^2 - \square}(C - m^2),$$

and therefore

$$\boxed{\kappa = \frac{g_H^2}{M^2} > 0.}$$

The sign is compatible with exchange of a positive-residue massive state and with standard positivity reasoning [Adams2006].

Model-specific tree-level sum rules

Parallel and perpendicular speeds satisfy

$$c_{\parallel}^2 = \frac{1 - \epsilon}{1 + \epsilon}, \quad c_{\perp}^2 = \frac{1}{1 + \epsilon},$$

hence

$$\boxed{c_{\parallel}^2 = 2c_{\perp}^2 - 1.}$$

For the minimal one-mediator completion, the k^4 dispersion coefficients also obey

$$\boxed{\alpha_{\parallel} = 4\alpha_{\perp}.}$$

These are not universal laws of two-fluid physics. They are consistency relations of the minimal model and will be perturbed by additional operators, resonances, and loops.

9. Gravity coupling and circularity

The direct substitution

$$g_{\mu\nu}^{\text{physical}} = C g_{\mu\nu}$$

is singular and mimetic. It does not evade the noninvertibility or extra-mode problem. A regular alternative introduces a Weyl compensator χ :

$$g_{\mu\nu} \rightarrow \Omega^2 g_{\mu\nu}, \quad \chi \rightarrow \Omega^{-1} \chi,$$

$$\hat{g}_{\mu\nu} = \frac{\chi^2}{6M_{\text{p}}^2} g_{\mu\nu}.$$

A representative action is

$$S_W = \int \sqrt{-g} \left[\frac{\chi^2}{12} R + \frac{1}{2} (\nabla \chi)^2 - \frac{f^2 \chi^2}{2} (Y_+ + Y_-) + \frac{\kappa}{2} (C - \zeta \chi^2)^2 \right].$$

On the locking surface, the phase state calibrates the same invariant metric used by gravity. This is a regular EFT representation of chronometric calibration. It does not derive Einstein dynamics from null phases. The conformal manifold and gravitational action remain inputs.